Confined Space Entry & Gas Testing Risk Assessment

Confined Space Entry Risk Assessment & Atmospheric Hold Points

Scope: Utility Chamber & Vessel Inspection - Atmospheric Limits: O2 19.5%-23.5%, LEL < 10%, H2S = 0 ppm

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01. Mechanical & Process Isolation
Atmospheric / Physical Hazard: Inadvertent inflow of toxic gas, steam, or liquid
Init Risk: 25 (Crit)
Controls & Hold Point: Positive spade/blinding + LOTO padlock register [HOLD POINT]
Res: 4 (Low)

02. Pre-Entry 4-Gas Testing
Atmospheric / Physical Hazard: Oxygen depletion / flammable vapor pocket
Init Risk: 20 (High)
Controls & Hold Point: Stratified top/mid/bottom probe test by Authorized Gas Tester
Res: 4 (Low)

03. Forced Air Ventilation & Entry
Atmospheric / Physical Hazard: Heat stress / toxic fume accumulation during work
Init Risk: 15 (High)
Controls & Hold Point: Continuous blower air changes + personal 4-gas monitor on entrant
Res: 3 (Low)

04. Standby Attendant & Rescue
Atmospheric / Physical Hazard: Unassisted entrant incapacitation inside chamber
Init Risk: 16 (High)
Controls & Hold Point: Dedicated hole-watch log + certified tripod winch & SCBA on standby
Res: 3 (Low)

Matrix - Entry & Atmosphere

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Site Access & Security
Hazard & Consequence: Unauthorised entry / public interface
Init Risk: 12 (Med)
Controls & Hold Points: Turnstile or gate log, visitor induction, hoarding integrity checks
Res: 3 (Low)

Temporary Electrics - Distribution
Hazard & Consequence: Damaged leads / RCD failure
Init Risk: 15 (High)
Controls & Hold Points: 110V or RCD-protected supplies, weekly PAT/inspection register, cable ramps
Res: 4 (Low)

Hot Works - Welding & Cutting
Hazard & Consequence: Fire / sparks to combustibles
Init Risk: 20 (High)
Controls & Hold Points: Hot work permit, fire watch during + 60 min after, extinguishers at point of work [HOLD POINT]
Res: 5 (Low)

Work at Height - Scaffold Use
Hazard & Consequence: Fall of person / incomplete scaffold
Init Risk: 25 (High)
Controls & Hold Points: Scafftag system, weekly + after-alteration inspections, no unauthorised alterations [HOLD POINT]
Res: 5 (Low)

Matrix - Work Inside the Space

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Lifting - Mobile Crane General
Hazard & Consequence: Overload / load swing
Init Risk: 20 (High)
Controls & Hold Points: Lift plan per lift category, certified gear register, anemometer monitoring [HOLD POINT]
Res: 4 (Low)

Plant Refuelling
Hazard & Consequence: Fuel spill / fire
Init Risk: 12 (Med)
Controls & Hold Points: Bunded bowser, drip trays, spill kit at point of use, no smoking zone
Res: 3 (Low)

Night Works
Hazard & Consequence: Poor visibility / fatigue
Init Risk: 15 (High)
Controls & Hold Points: Lighting survey (min lux at task), fatigue roster limits, hi-vis Class 3
Res: 5 (Low)

Heat Stress Management
Hazard & Consequence: Heat exhaustion / heat stroke
Init Risk: 16 (High)
Controls & Hold Points: WBGT monitoring, shaded rest + cool water, acclimatisation schedule, buddy checks
Res: 4 (Low)

Atmospheric Testing Log (Template)

Continuous 4-gas monitoring with logged readings at defined intervals. Alarm = evacuate immediately.

Oxygen (O-)
Safe Range: 19.5% - 23.5%
Alarm Action: Evacuate + ventilate
Frequency: Continuous

LEL (Flammables)
Safe Range: < 10% LEL
Alarm Action: Evacuate, no re-entry until cleared
Frequency: Continuous

H-S
Safe Range: < 10 ppm TWA
Alarm Action: Evacuate + respiratory protection review
Frequency: Continuous

CO
Safe Range: < 25 ppm TWA
Alarm Action: Evacuate + source investigation
Frequency: Continuous

Entry Control & Attendant Duties

The attendant never enters the space and never leaves the entry point while entrants are inside.

Entry Permit
Detail: Issued per entry, per shift
Control / Requirement: Permit displayed at entry point [HOLD POINT]
Verification: Permit board

Attendant Log
Detail: Entrant names, in/out times, comms checks
Control / Requirement: Log updated in real time
Verification: Entry log

Retrieval System
Detail: Tripod + winch rigged before entry
Control / Requirement: Non-entry rescue is the default method
Verification: Pre-entry check

Ventilation
Detail: Forced air, intake from clean zone
Control / Requirement: Minimum air changes per assessment
Verification: Flow check

Rescue Plan Summary

Rescue arrangements are in place and briefed BEFORE entry - improvisation is not a rescue plan.

Primary Method
Detail: Non-entry retrieval (winch/harness)
Control / Requirement: Tested at shift start
Verification: Drill record

Entry Rescue (Last Resort)
Detail: Trained team with breathing apparatus only
Control / Requirement: BA sets at entry point
Verification: Competency certs

Communications
Detail: Radio check entrant - attendant - supervisor
Control / Requirement: Backup signal agreed (line tugs)
Verification: Comms log

Drills
Detail: Simulated rescue per project phase
Control / Requirement: Timed + debriefed
Verification: Drill report

Matrix - Isolation & LOTO

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Earthworks - Bulk Excavation
Hazard & Consequence: Plant-pedestrian interface / reversing plant strike
Init Risk: 20 (High)
Controls & Hold Points: Trained banksman, segregated walkways, reversing cameras [HOLD POINT: daily plant check]
Res: 5 (Low)

Earthworks - Trench Support
Hazard & Consequence: Trench wall collapse / engulfment
Init Risk: 25 (High)
Controls & Hold Points: Batterered or shored sides per soil class, ladder access every 15 m, daily competent-person inspection [HOLD POINT]
Res: 5 (Low)

Formwork & Falsework Erection
Hazard & Consequence: Falsework overload / premature striking
Init Risk: 20 (High)
Controls & Hold Points: Designed falsework with tag system, striking only on engineer permit-to-strike [HOLD POINT]
Res: 4 (Low)

Reinforcement Fixing
Hazard & Consequence: Manual handling strain / protruding bar impalement
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical lifting for bundles, mushroom caps on starter bars, team-lift limits
Res: 4 (Low)

Matrix - Ventilation & Access

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Concreting - Pump & Skip
Hazard & Consequence: Hose whip / concrete splash to eyes
Init Risk: 15 (High)
Controls & Hold Points: Secured hose couplings with whip checks, sealed goggles, exclusion under skip path
Res: 5 (Low)

Blockwork & Plastering
Hazard & Consequence: Falling blocks / silica dust inhalation
Init Risk: 12 (Med)
Controls & Hold Points: Toe boards on scaffolds, wet cutting + FFP3 respirators, dust extraction
Res: 4 (Low)

Structural Steel - Bolt-up
Hazard & Consequence: Dropped tools / falling objects
Init Risk: 20 (High)
Controls & Hold Points: Tool lanyards, bolt bags, hard-barricaded drop zone below [HOLD POINT: wind limit 20 kt]
Res: 5 (Low)

Roofing Works
Hazard & Consequence: Fall through fragile surface
Init Risk: 25 (High)
Controls & Hold Points: Crawling boards, edge protection, harness anchor plan [HOLD POINT: fragile surface permit]
Res: 5 (Low)

Matrix - Monitoring & Housekeeping

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Dust-Generating Works
Hazard & Consequence: Respirable crystalline silica
Init Risk: 16 (High)
Controls & Hold Points: Water suppression, on-tool extraction, FFP3 face-fit tested, health surveillance
Res: 4 (Low)

Noise-Intensive Works
Hazard & Consequence: Noise-induced hearing loss
Init Risk: 12 (Med)
Controls & Hold Points: Noise zones signed, hearing protection mandatory >85 dB(A), plant selection low-noise first
Res: 3 (Low)

Traffic Management - Internal Roads
Hazard & Consequence: Vehicle collision at pinch points
Init Risk: 16 (High)
Controls & Hold Points: One-way system, 10 km/h limit, passing bays, trained traffic marshals
Res: 4 (Low)

Welfare - Water, Rest, Sanitation
Hazard & Consequence: Dehydration / hygiene illness
Init Risk: 9 (Med)
Controls & Hold Points: Potable water points, shaded rest areas, serviced sanitation per headcount ratio
Res: 2 (Low)

Matrix - Monitoring & Housekeeping

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Adverse Weather - Lightning & Storm
Hazard & Consequence: Lightning strike / crane instability
Init Risk: 15 (High)
Controls & Hold Points: 30-30 lightning rule, crane stand-down limits, secure loose materials
Res: 5 (Low)

Housekeeping - Access Routes
Hazard & Consequence: Slips, trips and falls (same level)
Init Risk: 12 (Med)
Controls & Hold Points: Designated walkways kept clear, cable covers, daily supervisor walk-round
Res: 3 (Low)

Visitor & Delivery Driver Safety
Hazard & Consequence: Unfamiliar persons in live work zones
Init Risk: 12 (Med)
Controls & Hold Points: Escorted access only, driver briefing card, PPE issue at gate
Res: 3 (Low)

Document Control & Record Retention
Hazard & Consequence: Superseded revision used in field
Init Risk: 9 (Med)
Controls & Hold Points: Controlled copy register, revision check at TBT, obsolete copies withdrawn
Res: 2 (Low)

Activity Matrix - Continuation Part 2

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Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Incident Reporting & Escalation
Hazard & Consequence: Delayed reporting / evidence loss
Init Risk: 12 (Med)
Controls & Hold Points: Immediate verbal + 24h written report, scene preservation, investigation per severity matrix
Res: 3 (Low)

Subcontractor Coordination (SIMOPS)
Hazard & Consequence: Conflicting simultaneous operations
Init Risk: 16 (High)
Controls & Hold Points: Daily SIMOPS meeting, shared permit board, interface risk review [HOLD POINT]
Res: 4 (Low)

Manual Handling - General
Hazard & Consequence: Musculoskeletal injury
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical aids first, load weight labels, manual handling training records
Res: 3 (Low)

Storage of Flammables & Gas Cylinders
Hazard & Consequence: Fire / cylinder projectile
Init Risk: 16 (High)
Controls & Hold Points: Ventilated cage, upright chained cylinders, separation distances, signage
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Excavation Services - Trial Pits
Hazard & Consequence: Unknown service strike
Init Risk: 20 (High)
Controls & Hold Points: Drawings + CAT scan + hand-dig only within 500 mm of marked services [HOLD POINT]
Res: 5 (Low)

Concrete Washout
Hazard & Consequence: Alkaline runoff to ground/drain
Init Risk: 9 (Med)
Controls & Hold Points: Designated lined washout pit only, pH check before disposal
Res: 2 (Low)

Tower Crane - Climbing Operations
Hazard & Consequence: Structural overload during climb
Init Risk: 25 (High)
Controls & Hold Points: OEM climbing procedure, specialist crew, wind hold limits [HOLD POINT: climb permit]
Res: 5 (Low)

Demolition - Soft Strip
Hazard & Consequence: Hidden asbestos / structural instability
Init Risk: 20 (High)
Controls & Hold Points: Pre-demolition survey, licensed removal where identified, engineer sequence sign-off [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Piling - Bored Operations
Hazard & Consequence: Auger entanglement / bore collapse
Init Risk: 20 (High)
Controls & Hold Points: Auger guards, exclusion zone, bore capping same shift
Res: 5 (Low)

Roadworks Interface - Public Highway
Hazard & Consequence: Public vehicle incursion
Init Risk: 20 (High)
Controls & Hold Points: Approved traffic management layout, advance signage, barrier-rated separation [HOLD POINT]
Res: 5 (Low)

Confined Welfare - Drinking Water Quality
Hazard & Consequence: Contaminated supply illness
Init Risk: 9 (Med)
Controls & Hold Points: Tested potable supply, dated containers, cleaning schedule
Res: 2 (Low)

Fatigue - Extended Shifts
Hazard & Consequence: Error from exhaustion on safety-critical tasks
Init Risk: 15 (High)
Controls & Hold Points: Max shift + rest rules, no safety-critical solo work after extended hours
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Young & Inexperienced Workers
Hazard & Consequence: Unrecognised hazards
Init Risk: 12 (Med)
Controls & Hold Points: Named supervisor, restricted task list, enhanced supervision ratios
Res: 3 (Low)

Emergency Drill Programme
Hazard & Consequence: Unpractised evacuation / slow muster
Init Risk: 12 (Med)
Controls & Hold Points: Monthly drill rotation (fire, rescue, spill), timed muster with roll call
Res: 3 (Low)

First Aid Provision
Hazard & Consequence: Inadequate response to injury
Init Risk: 12 (Med)
Controls & Hold Points: Trained first aiders per shift ratio, stocked stations, hospital route posted
Res: 3 (Low)

PPE Issue & Compliance
Hazard & Consequence: Non-use / damaged PPE
Init Risk: 9 (Med)
Controls & Hold Points: Free issue + fit checks, PPE matrix by task, supervisor spot checks
Res: 2 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Safety Signage & Barriers Audit
Hazard & Consequence: Missing or ignored warnings
Init Risk: 9 (Med)
Controls & Hold Points: Weekly audit walk, multilingual pictorial signage, barrier integrity checks
Res: 2 (Low)

Toolbox Talk Programme
Hazard & Consequence: Generic talks with no task relevance
Init Risk: 9 (Med)
Controls & Hold Points: Task-specific TBT before each shift, attendance signed, topics logged weekly
Res: 2 (Low)

Management Site Tours
Hazard & Consequence: Hazards normalised at worker level
Init Risk: 9 (Med)
Controls & Hold Points: Weekly leadership tour with close-out actions tracked to completion
Res: 2 (Low)

Change Management - Scope Variation
Hazard & Consequence: Unassessed new hazards from changed method
Init Risk: 16 (High)
Controls & Hold Points: MOC form before variation starts, RA revision re-briefed to crew [HOLD POINT]
Res: 4 (Low)

Confined Space Entry & Gas Testing Risk Assessment

Residual Risk Summary & ALARP Statement

All activities are reduced to Low/Medium residual risk once the stated controls are verified in place. No activity may start on initial risk alone.

High (15-25) initial
Count of Activities: Assessed per activity matrix
Action Rule: Controls + hold points mandatory before start
Authority: Site HSE authority

Medium (8-12) residual
Count of Activities: Monitor at supervision level
Action Rule: Daily TBT reinforcement + spot checks
Authority: Supervisor

Low (1-6) residual
Count of Activities: Majority of activities after controls
Action Rule: Proceed under permit and supervision
Authority: Permit issuer

ALARP Statement
Count of Activities: Further risk reduction considered for every High initial item
Action Rule: Grossly disproportionate measures documented, not silently dropped
Authority: HSE Officer

Hold Point Register

Work must stop at each hold point until the named authority verifies and releases it. Hold points are never self-released.

HP-01 Permit Verification
Where It Applies: Start of each high-risk activity
Released By: Permit issuer
Evidence: Signed permit at workface

HP-02 Isolation / LOTO
Where It Applies: Energy isolation before work
Released By: Authorised isolator
Evidence: Lock register + prove-dead test

HP-03 Pre-Start Inspection
Where It Applies: Plant, scaffold, lifting gear
Released By: Competent inspector
Evidence: Tag / inspection record

HP-04 Authority Sign-Off
Where It Applies: Document approval before execution
Released By: Client/site HSE authority
Evidence: Signed approval block

Review, Approval & Revision Record

This document is a professional draft. Field use requires review and formal approval by the client/site HSE authority.

Prepared By
Name / Organisation: SafetyWithAthar (remote)
Responsibility: Draft preparation from supplied work-pack inputs
Status: Rev. 0 issued

Reviewed By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Site HSE review against actual conditions
Status: [HOLD POINT]

Approved By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Client / site HSE authority approval
Status: [HOLD POINT]

Revision History
Name / Organisation: Rev. 0 - first issue for review
Responsibility: Superseded revisions withdrawn from field use
Status: Controlled